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Self-Healing Databases for Emergency Response Logistics in Remote and Infrastructure-Poor Settings

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Abstract: *Background:* Accurate, real-time data about response technologies, capabilities, and availabilities are key to effective emergency response logistics; this is particularly important in remote settings, such as in the Arctic, where limited infrastructure, logistics, and technologies occasion the need for careful planning and immediate response in a fragile, pristine, and rapidly changing ecosystem. Despite persistent calls for improved data quality, processing, and analysis capabilities to support Arctic emergency response logistics, these issues have not been addressed and advanced analytical methods available in other safety-critical and oil and gas settings, such as machine learning, artificial intelligence (AI), or emergent, self-aware, and self-healing databases, have not been widely adopted. *Methods:* This work explores this research gap by presenting a machine learning algorithm and self-healing database approach, describing its application in Arctic logistics and emergency response. *Results:* The self-healing algorithm could be applied to other safety-critical databases that could benefit from technology that automatically detects, diagnoses, and repairs data anomalies and inconsistencies, with or without human intervention. *Conclusions:* The results show significant improvements in data cleaning and analysis, and for emergency response logistics data, planning, and analysis, along with future research and research needs in remote and infrastructure-poor settings.

Keywords: artificial intelligence; logistics planning; data quality; machine learning; self-healing databases; emergency response

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1. Introduction

Climate change, energy and natural resource developments, and thinning Arctic sea ice have increased the accessibility of Arctic waters for energy exploration and development [1], fishing [2], tourism [3], and maritime shipping [4]. Over the past decade, vessel transits through the Bering Strait have increased by 88%, from 262 in 2009 to 494 in 2019 [5]. At the same time, transits through the Northern Sea Route (NSR), to and from Russia across Norway, increased by 28% between 2019 and 2021 [6]. Ship traffic is expected to increase over the next decade and beyond, as the NSR becomes a more attractive shipping route for vessels traveling between Western Europe and Asia along the Russian coast, and offshore oil development in the Arctic Ocean; onshore oil, gas, and mining operations in Alaska; and cruise vessel traffic are expected to grow [5].

Just-in-time expectations for goods and services have increased global pressure for shorter shipping timelines, for faster turn-rounds, and for increased shipping activity in the Arctic, perhaps with different attention to safety risks. In September 2024, the first

large Panamax container ship transited from Europe to Shanghai in 16 days, which saved days and weeks of Suez Canal transit [7]. A month later, in October 2024, a Chinese vessel without an ice class certification traveled across the Northern Sea Route with sanctioned goods for Russia's Arctic LNG 2 project [8]. As pressures and expectations mount with increased Arctic shipping, so too does the potential for a major Arctic oil spill, which could have enormous impacts on the region's economy, ecosystems, social and cultural resources, and fragile subsistence communities [9,10]. Nations, states and provinces, industries, and non-profit enterprises have significantly invested in emergency response logistics planning to address these risks, through systemic and localized risk assessments [11], resource allocation modeling [12], studies of oil and gas characteristics and toxicity [13], the development of Arctic transport decision support systems [14], and studies of the impacts of oil on wildlife and marine biota, communities, and ecosystems, particularly in ice and extreme conditions [15].

Key to effective emergency response logistics are accurate, real-time data about technologies, capabilities, and availabilities; this is particularly important in remote settings, such as in the Arctic, where limited infrastructure, logistics, and technologies occasion the need for careful planning and immediate response in a fragile, pristine, and rapidly changing ecosystem [16–18]. In the Arctic, the lack of information, communications, and transportation infrastructure, and the enormous distances between sparsely populated villages, coupled with the unpredictability of weather, and the magnitude and power of sea states and environmental conditions, complicate access to information, and can delay deployment and supply of critical resources. Public expectations for four-season response capabilities in the event of an incident in the Arctic also increase the need for timely analysis of reliable equipment, personnel, and resource information to support effective Arctic emergency response logistics [19].

Data issues in Arctic emergencies and oil spill response have long been studied, with many calls for improved data quality, availability, and integration [9,10,20–22]. Emergency response logistics data are often disaggregated, incomplete, and captured by multiple stakeholders using different platforms, standards, and data definitions [9,22]. Disaggregated and non-standard data in logistics and emergency response events, which involve federal, state and local agencies, industries, and environmental and cultural stakeholders, can result in delayed response activities, driven by a lack of shared situational awareness, and data integrity and compatibility issues [9,20,21]. Most recently, the U.S. Government Accountability Office called for improvements in Coast Guard operational data in the Arctic, citing incomplete and unavailable data, reports, and performance metrics [23].

Despite persistent calls for improved data quality, processing, and analysis capabilities to support Arctic emergency response logistics, these issues have not been addressed and advanced analytical methods available in other safety-critical and oil and gas settings, such as machine learning, artificial intelligence (AI), or emergent, self-aware, and self-healing databases [24,25], have not been widely adopted. This is a critical research gap in remote and infrastructure-poor logistics settings such as Arctic emergency response, where the need for real-time response is critical and the human capital resources to effect response, and to maintain safety-critical databases, are sparse. After describing previous work, this paper addresses that research gap by presenting a machine learning algorithm and self-healing database approach, and describing its application in Arctic logistics and emergency response. The self-healing algorithm could be applied to other logistics, emergency response, and safety-critical databases that could benefit from technology that automatically detects, diagnoses, and repairs data anomalies and inconsistencies, with or without human intervention. The results suggest improvements for emergency response logistics data,

planning, and analysis, along with future research and research needs, particularly in remote and infrastructure-poor settings.

2. Previous Work

Effective emergency response and logistics operations require a confluence of well-orchestrated activities: efficient resource allocation, shared situational awareness of the response effort, and efficacious response [26,27]. Central to the logistics of resource allocation, situation awareness, and effective response are accurate, complete, and available data about personnel, equipment, and funding availability and capabilities, and the ability to develop response strategies in real-time. Optimized response in resource-constrained settings such as the Arctic also require attention to information about events, response timelines, resource transit times, and logistics, as well as environmental and social considerations, such as wildlife and marine mammal impacts [28], cultural concerns [29], oil and fuel characteristics, and behavior in extreme settings, and impacts on local and subsistence communities [22,30].

Emergency response logistics thus require high-quality, accurate, and complete data about transportation, personnel, routes, weather, events, and other contingencies, as well as information about response equipment, resources, and personnel. Information about events includes location, timing, duration, severity, trends, or change data, as well as information about resources deployed, their timing, and effectiveness. Resources have costs, locations, shelf lives, optimal deployment periods, transit times, constraints about their use and maintenance, as well as weight, height, transportation, and storage considerations, knowledge of which is critical for effective response and mission completion. In the Arctic, emergency response logistics also rely on current environmental information such as weather, ice, and storm impact data, as well as information about community, marine, and subsistence economy information, including wildlife, sea mammal and bird migration, and hunting seasons and patterns, all of which can impact an effective response [9].

Data analysis and modeling also rely on common data definitions, consistent data structures, and data capture standards, especially since participants in Arctic logistics and emergency response are geographically scattered, with varying degrees of access to and understanding of the on-the-ground operational picture. Distributed decision-makers in large-scale safety–critical systems often rely on standard data definitions and classifications to ensure that all members are using the same data to develop shared mental models of the system and to make decisions [31]. Standard data definitions and ontologies help ensure that data have been captured, represented, labeled, and stored with consistent data descriptions and nomenclature, that different datasets can be shared and integrated, and that data dictionaries represent and store the same data. Absent standard data definitions and ontologies, and classification schemes, decision-makers, regulators, analysts, and modelers can face difficulties relying on critical data, and significant data cleansing, reconciliation, and management efforts may be required.

Real-time analysis of emergency response logistics data is also important so that data sources, analytical tools, and methods for assessing data and identifying trends reflect current conditions and response needs. However, real-time data analysis for Arctic emergency response logistics is complicated by long-standing data challenges [9,10,12]. Improved analytic processes, including machine learning applications for global supply chain enhancement, have been developed [32], but applications in Arctic emergency response logistics have been needed for some time, stemming from pressure for real-time analysis of emergency response data, and for real-time predictive models for risk analysis, resource allocation and planning. Machine learning applications integrated with AI and the Internet

of Things (IoT) are beginning to be introduced, linking data from assets, personnel, and systems, and increasing expectations for real-time analysis [30].

The power of real-time machine learning and other analytical approaches has often been limited; however, in cases where data are inconsistent, missing, or inaccurate [33]. As a result, interest developed in automated and autonomous capabilities to detect system anomalies, diagnose solutions, and, with or without human intervention, correct anomalies [34]. These systems, which may utilize embedded neural nets or other machine learning techniques, have shown improved database accuracy and integrity [35], correcting faults automatically and, optimizing query performance through machine learning, AI, and deep learning models [36], and performing self-maintenance [37].

Self-healing systems are one form of autonomic computing capabilities, which also include self-configuration, self-optimization, and self-protection [38]. Self-healing systems detect and can address system and data anomalies, integrating with machine learning models and learning from historical data, real-time analytics, and predictive models [39]. The integration of self-healing databases with AI, machine learning, and deep learning offers the potential for anomaly detection, self-optimization, and automated recovery in databases without human intervention [36,37]. Despite these advances in other domains, however, self-healing systems have not been utilized extensively in emergency response logistics, although their potential has been recognized [40].

Emergent systems are another form of learning technology that mimics self-organization principles and capabilities in social and organizational systems [41] and offers capabilities for self-configuration, optimization, healing, and protection [42]. Pattern recognition capabilities in emergent systems can be used to identify anomalies in behavior, movement, analysis, or data, often using sequences that match specific patterns and suggesting novel solutions [43]. AI-driven anomaly detection for self-optimizing systems has been used in cloud management systems to analyze patterns, offer predictions, or enhance system performance [44], and applications in emergency response and logistics planning are emerging [45,46].

Technology approaches and solutions to address long-standing and persistent data, processing, and analysis issues in emergency response logistics data, therefore, exist; however, they have not been developed or applied in remote and infrastructure-poor logistics settings such as the Arctic emergency response, where the need for a real-time response is critical and the human capital resources to effect response, and to maintain safety-critical databases, are sparse. This paper describes one such approach and relates experience with a neural network-based approach for anomaly detection, diagnosis, and self-healing in emergency response logistics databases, with attention to the limits and complexity they introduce in real-time analysis in safety-critical systems, particularly in remote locations [44]. The following section describes an Arctic logistics and emergency response case study with challenges in missing, incomplete, and redundant data in an infrastructure-poor setting where real-time analysis and logistics planning are required.

3. Materials and Methods

Timely analysis of oil spill prevention and emergency response logistics information is particularly important in remote settings such as the Arctic, as the lack of Arctic infrastructure and the need for careful planning and immediate response makes operations difficult in this fragile, pristine and rapidly changing ecosystem with extended logistical supply chains, personnel shortages, and long distances to travel [9–11]. Despite the need for advanced analytics and machine learning techniques to optimize and improve logistics and emergency response in the Arctic, these approaches have not been applied in this setting. Previous work identified the need for high-quality, accurate, and complete data, common

data standards, and real-time logistics and emergency response analysis. However, little work has developed advanced analytical approaches to address these challenges, including applications of machine learning and self-healing capabilities that could improve logistics and emergency response analyses and processes, and surmount human capital challenges in remote and infrastructure-poor settings. This paper presents such an approach and describes its application in Arctic logistics and emergency response. The self-healing technique addresses the need for improved data quality and processing for real-time emergency response and logistics in an environmentally sensitive and safety-critical system with limited physical, technological, and human resources.

3.1. Setting and Background

The setting for this study is the U.S. Arctic and Western Alaska, where vessels transit the Gulf of Alaska, the Bering Sea and Bering Strait, the Aleutian Islands, and along the Western coast of Alaska, through the Chukchi and Beaufort Seas (Figure 1).

Figure 1. U.S. Coast Guard Arctic Area of Responsibility [23].

As Arctic waters warm, more vessels are engaged in Arctic voyages, particularly as the prospect of an ice-free passage through the Northern Sea Route across Russia or the Northwest Passage across Canada looms [5]. To protect the Arctic and other waters, and to support safe and secure maritime transportation, the U.S. Coast Guard specifies oil spill response requirements for owners or operators of regulated tank and non-tank vessels when in U.S. waters. These requirements are codified in the Oil Pollution Act of 1990 (OPA-90) [9,47], which was passed by the U.S. Congress following the 1989 *Exxon Valdez* oil spill in Prince William Sound, Alaska [47,48]. For Arctic vessel transits, the state of Alaska also prescribes vessel operations, maintenance, and oil spill response planning and equipment requirements for vessels that operate within three nautical miles of the shore and that call on ports in Alaska [49].

Vessels operating in the U.S. Arctic and Western Alaska must document their required oil spill response resources and plans in Vessel Response Plans (VRPs), which are guided by international standards and rules. In addition to specifying what equipment (containment mechanisms, deployment tools, etc.) needs to be available, VRPs also describe oil spill response logistics best practices, guidelines for vessel outfitting and equipment, requirements

for oil spill response drills, fuel sulfur limits, and maritime transit areas to be avoided. Federal laws and regulations also require VRPs to identify contracts for oil spill response services, salvage, and qualified oil spill response individuals. These requirements are linked to a vessel's worst-case discharge (WCD) metric, the volume of oil that could be discharged from the vessel in adverse weather (U.S. 33 Code of Federal Regulations (CFR) 155.1020).

3.2. Approach

In this study, data were collected by our research partner, an Oil Spill Removal Organization (OSRO), as part of their oil spill response monitoring and data collection activities. The dataset was cleaned using automated scripts to reduce duplicative and conflicting data records. Following the automated scripts data cleaning, the process was repeated on the same dataset with a self-healing process using machine learning. Process improvement metrics were then calculated, and observations and insights were made regarding the use of self-healing and emergent approaches to address data quality issues for Arctic oil spill response and logistics analysis.

3.3. Data

The data comprised 11,568 oil spill response records from 2048 participating vessels making transits through waters off Western Alaska and the Arctic between 1 January and 31 August 2023. The attributes in the dataset are shown in Table 1, and the vessels and vessel data types are shown in Table 2.

Table 1. Data attributes.

Data Identification	Data Attributes
1	Vessel ID (MMIC)
2	Vessel Name
3	Last port of call (LPOC)
4	Next port of call (NPOC)
5	Intended route
6	Estimated date of arrival
7	Estimated time of arrival
8	Fuel oil type
9	Fuel oil quantity
10	Lube oil type
11	Lube oil quantity
12	Cargo on board (type)
13	Cargo volume
14	Location of last fuel received
15	Exhaust scrubber installed? (yes/no)
16	Vessel contact information—Email
17	Vessel contact information—Phone
18	Intended route deviating from routing measures? (yes/no)
19	On board Automated Identification System functioning? (yes/no)
20	On board Automated Identification System tested? (yes/no)
21	On board Automated Identification System date tested

Table 2 shows the different types of vessels in the database; there were no records for break bulk, ferry, fishing, heavy lift, oil, oil chemical or miscellaneous vessels in the dataset. Incomplete data about one pleasure craft/yacht and two research/survey vessels were deleted from the dataset. There were no 'blank' fields for vessel type.

Table 2. Number and types of vessels.

Vessel Type	Number of Vessels (2023)	Percentage
Break Bulk	5	0.2%
Bulk Carrier	981	47.9%
Bulk/Container	59	2.9%
Container	684	33.4%
Heavy Lift	3	0.1%
LNG/LPG	36	1.8%
Passenger	85	4.2%
Refrigerated Cargo	36	1.8%
RoRo	9	0.4%
Tug/Offshore Supply Vessel	1	0.0%
Vehicle Carrier	149	7.3%
Total	2048	

3.4. Method

Visual Basic (VBA) scripts using the Excel macros shown in Appendix A were developed to organize and clean the original dataset. The baseline scripts were reviewed using a generative AI tool, ChatGPT; the reviewed scripts were then modified to provide a context for further development (red, green, and blue color-coded data for certain data criteria (e.g., missing data)), and script instructions were developed to handle the resulting dataset (e.g., condense the multiple color-coded rows to one row with multiple columns). Once the missing data scripts were finalized, additional scripts to eliminate duplicate data were then created organically using lessons learned from the missing data script development. The scripts were tested with an open-source dataset from an academic setting; no oil spill response data were ingested into a generative AI tool. The tested scripts were then used to clean a copy of the oil spill response dataset, by moving data in color-coded cells to green (cleaned cells), eliminating duplicate information. The resulting database was then reviewed by visual inspection to ensure that all missing and duplicative data had been flagged, captured, and resolved. The Visual Basic scripts are provided in Appendix A.

The automated analysis showed that, of the original 11,568 records, 6490 (56.1%) were duplicative, and 118 were missing transit exit dates or vessel type data (1.02%). Other incomplete records were then identified and deleted from the database, which resulted in an additional reduction of 2891 records (an additional 24.99% reduction). The inspection also showed that three data points skewed the data distribution; these outlier data were reviewed with the research partner, confirmed to be incorrect, and were removed from the dataset. The dataset resulting from the automated cleaning resulted in 2069 records, which was 17.89% of the original dataset (Table 3).

Table 3. Automated data cleaning results—Oil Spill Response Organization Records, 2023.

Types of Records	Number of Records	Percentage of Original Dataset
Initial number of records	11,568	
Duplicative records	6490	56.10%
Incomplete records	118	1.02%
Incomplete oil records	2891	24.99%
Final dataset *(eliminating duplicate, incomplete records)*	2069	17.89%

4. Analysis and Results

The initial database analysis using automated anomaly detection scripts highlighted a number of data difficulties. Incompatible data capture methodologies, data standards,

and data and metadata definitions have been emblematic of emergency and oil spill response data challenges for many years [9,50]. In this case study, incompatible data capture policies and standards led to the development of a database that required substantial cleaning prior to analysis. Data cleaning has long been highlighted as a bottleneck in data analytics [51]; in emergency response logistics data analysis, the lack of common standards and definitions have resulted in data quality issues with a significant impact on data accuracy and usability [20].

Interest in improving the data analysis process arose following the automated scripts process, which, although automated, still required significant time for anomaly detection and data cleaning. As a result, neural network techniques were applied for data cleaning with the original dataset. A neural network model was created using freely available open-source resources (Appendix B). A preliminary algorithm was developed that used color coding to highlight duplicative and incomplete data. The results were then reviewed by a human subject matter expert. The algorithm was then updated using a second set of training data from the original dataset, and was run again on the large dataset. Redundant data were deleted, and a tracer record was created for audit and compliance testing and analysis. Missing and incomplete data were flagged by the neural net so that the dataset could be updated with the missing information. Machine learning was used to propose potential data to complete the data records based on contextual information and data in the dataset. Process improvements in terms of time differences were recorded, comparing the use of the automated scripts and the neural nets algorithm. The results are shown in Table 4.

Table 4. Database analysis process.

Phases	Automated Anomaly Detection Hours *(Scripts)*	Self Healing Hours *(Neural Nets)*	Process Improvement *(% Change in Hours)*
Data Collection	0	0	-
Data Cleaning	120	26	78.33%
Data Analysis	16	16	-
Total	136	42	69.1%

Interest in improving the data analysis process arose following the automated scripts process, which, although automated, still required significant time for anomaly detection and data cleaning. As a result, neural network techniques using the Python code in Figure A5 in Appendix B were applied for data cleaning with the original dataset. A neural network model was created using freely available open-source resources. A preliminary algorithm was developed that used color coding to highlight duplicative and incomplete data. The results were then reviewed by a human subject matter expert. The algorithm was then updated using a second set of training data from the original dataset, and was run again on the large dataset. Redundant data were deleted, and a tracer record was created for audit and compliance testing and analysis. Missing and incomplete data were flagged by the neural net so that the dataset could be updated with the missing information. Machine learning was used to propose potential data to complete the data records, based on contextual information and data in the dataset. Process improvements in terms of time differences were recorded, comparing the use of the automated scripts and the neural nets algorithm. The Python code developed is shown in Figure A5 in Appendix B.

The differences in time required for data cleaning using the automated anomaly detection scripts and using the neural nets approaches were significant, with a 78.33% reduction in time with the neural nets (Table 4), which used Python code to accelerate the cleaning process. The Visual Basic script data cleaning required over three times the cleaning time compared to the use of the neural nets, showing significant process improvements.

5. Discussion

Real-time data analysis using current and accurate data is imperative in safety-critical settings, including logistics planning for emergency and oil spill response in the Arctic, processes which are time-sensitive, carried out in high-pressure settings, and are impacted by physical, technological, and human resource limits [52,53]. In this context, this research explored an approach to addressing the need for high-quality data, real-time data analysis, consistent data structures, and improved data analytic processes in an environmentally sensitive and safety-critical setting with limited physical, technological, and human resources. Previous research identified gaps in Arctic emergency and oil spill response data, logistics planning, and analysis [9,22,54], and suggested the role that advanced analytics and AI could play in advancing emergency response logistics planning and analysis [37,55]. As hardware, software, and analytic solution costs decrease over time, AI and machine learning solutions have become accessible and realistic possibilities, even in settings with limited infrastructure. Human resource limitations in terms of technological knowledge and availability compound challenges in Arctic emergency response logistics and planning, but those limits could be addressed with the advent of reliable, affordable, and sustainable machine learning support [14,21].

High-quality, complete, and consistent data are required for effective decision support in emergency response and safety settings, not only to ensure that decisions taken are informed by accurate data, but also in order to support timely and efficient information processing. This research utilized self-healing databases that embedded a machine learning classification algorithm in order to improve data analytics processes. The process improvements focused on data cleaning since those tasks require significant time and effort [51]. The neural net model deployed was able to detect, diagnose, and correct anomalies in the dataset, resulting in several improvements:

- *Improved data quality*, as missing, incomplete, and redundant data were identified and resolved using the neural net's classification algorithm;
- *Improved data analysis processes*, particularly with data cleaning tasks, compared to the use of automated anomaly detection scripts;
- *Improved support for real-time data anomaly and inconsistency detection*, as inconsistent, conflicting, and missing data were automatically flagged with the use of the classification algorithm;
- *Improved utilization of scarce human resources*, who could supervise and review the results of the automated data anomaly detection, diagnosis, and correction processes, rather than engage in time-consuming, cumbersome, and error-prone human data cleaning and review processes.

This research fills a gap in Arctic and remote logistics and emergency response systems, describing a self-healing machine learning algorithm that addresses the need for improved data quality and data processing for real-time emergency response and logistics in an environmentally sensitive and safety-critical system with limited physical, technological, and human resources. Previous work identified the need for high-quality, accurate, and complete data, common data standards, and real-time logistics and emergency response analysis. However, little work has developed advanced analytical approaches to address these challenges, including applications of machine learning and self-healing capabilities that could improve logistics and emergency response analyses and processes, and surmount human capital challenges in remote and infrastructure-poor settings. In addition, little work has developed these analytic techniques for real-time applications in these settings; thus, the novel contributions of this research are the development of machine learning and self-healing database capabilities, and their real-time application in a safety- and time-

critical setting, showing improved data quality, data analysis, and data processes, all critical goals in safety and logistics analyses.

Consistent data structures, data capture, and standards are integral to emergency response logistics planning [56,57]. However, Arctic emergency response logistics occur in a setting plagued by technological, data, and network infrastructure challenges [9]. Coast Guard data needs for Arctic missions, including search and rescue, emergency response, and oil spill prevention and response, have long been of concern [23], and maritime databases, such as the one utilized in this research, have similar issues. Central to maritime data quality issues are inconsistent data standards, metadata and data structures, and data capture processes [20]. The machine learning techniques employed in this research could be utilized to address the underlying and systemic issues related to maritime data quality, introducing self-healing databases in this setting. Yet to be resolved, however, are data integration and data management issues associated with linking self-healing databases and traditional databases [58], a challenge that was noted, but not addressed, in this research.

The use of machine learning and AI approaches in emergency logistics planning is still emerging, and interest in these tools and approaches is growing [59,60]. The use of AI-driven anomaly detection using algorithms and predictive models is increasing, including the potential for self-aware and self-healing databases [40]. However, these techniques introduce challenges with respect to integrating or embedding machine learning models with or in traditional emergency response logistics datasets. Such integration can lead to 'data friction,' when models and datasets developed using different standards and methodologies are linked, leading to potential disconnects in analysis, data, and data processing [61]. Integrating machine learning capabilities into existing emergency response logistics databases will make analyses more data-driven, with increasing needs for interoperability among data, tools, and models. Metadata—usually viewed simply as 'data about data'— will play an increasingly important role in interoperability, even if metadata may be a source of friction between differing models, tools, and datasets [62]. Improved data analytic processes, therefore, will likely occasion a need for continued attention to integration, interoperability, and the impacts of disparate datasets [23,62].

There are a number of limitations with this research that should be noted. First, the self-healing algorithms were developed with a single dataset comprising data from 2048 ships and multiple Arctic shippers for an 8-month period in 2023. The time period covered was limited, and future work should clearly extend the data collection longitudinally, and with more shipping companies, over a longer time period. The data were not validated against another dataset, and their representation across a diverse set of Arctic shipping companies, although wide, was not universal.

Data quality issues in emergency response logistics have been noted for many years. Historical criticisms include the lack of complete, accurate, and validated datasets that continue to the present day [10,23]. Developing and training machine learning and self-healing algorithms and databases on incomplete, missing, or conflicting data introduces challenges in algorithm development, creates non-standard data, and raises questions about the validity of the models, and their learning over time [63]. Dataset size also impacts the validity and reliability of machine learning results [64]; missing, incomplete, and redundant data contribute to reduced sample sizes for machine learning algorithms and other statistical analyses. Thus, caution should be exercised with the results of this study, and the generalization of these results outside of the Arctic and Western Alaska should be explored. Those facets not examined in this research were the root causes of the data quality issues, or the lack of systemic organizational response to address the underlying data quality issues [23]. Root cause analyses, perhaps employing AI, might provide important input to learning algorithms, leading to training and learning that could

address and rectify the human vulnerabilities and errors that led to the development of weak datasets [65].

In this study, missing and incomplete data were flagged by the self-healing algorithm so that the dataset could be updated with the missing information. Machine learning was used to propose potential data to complete the data records, based on contextual information and data in the dataset. Imputation and data replacement methods could also have been employed [66]. Redundant data were deleted, and a tracer record was created for audit and compliance testing and analysis. In future work, these steps could be automated, with or without appropriate human intervention, to further improve and accelerate the process of developing clean data. The strength of the findings of this research could be improved in the future with longitudinal analysis, using the improved or 'healed' data, which were beyond the scope of this initial research; insights into the process and decision support improvements could then be noted. Finally, this research was carried out in a technologically challenging and infrastructure-poor setting. Future work could address the impacts of technology, operational, and infrastructure constraints on advanced analytical work, including the development of self-healing databases.

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Appendix A

Visual Basic (VBA) scripts using the Excel macros shown in Appendix A were developed to organize and clean the original dataset. The baseline scripts were reviewed using a generative AI tool, ChatGPT; the reviewed scripts were then modified to provide a context for further development (red, green, and blue color-coded data for certain data criteria (e.g., missing data)), and script instructions were developed to handle the resulting dataset (e.g., condense the multiple color-coded rows to one row with multiple columns). Once the missing data scripts were finalized, additional scripts to eliminate duplicate data were then created organically using lessons learned from the missing data script development. The scripts were tested with an open-source dataset from an academic setting; no oil spill response data were ingested into a generative AI tool. The tested scripts were then used to clean a copy of the oil spill response dataset, by moving data in color-coded cells to green (cleaned cells), eliminating duplicate information. The resulting database was then reviewed by visual inspection to ensure that all missing and duplicative data had been flagged, captured, and resolved. Figures A1–A4 show the macros created to clean the data.

```
- Sub TransferUniqueMetricsToAboveGreenRow()
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets("Good ACN Data 1")

    Dim sourceRow As Long, lastRow As Long, greenRow As Long
    Dim cellValue As Variant
    Dim uniqueValues As Collection
    Set uniqueValues = New Collection

    Dim greenColor As Long, blueColor As Long, redColor As Long
    greenColor = 11854022
    blueColor = 15917529
    redColor = 11389944
    Dim colOffset As Integer

    lastRow = ws.Cells(ws.Rows.Count, "V").End(xlUp).Row

    Application.ScreenUpdating = False

    For sourceRow = 26 To lastRow
        greenRow = sourceRow - 1
        colOffset = 0


        If ws.Cells(greenRow, "V").Interior.Color = greenColor Then

            Do While ws.Cells(sourceRow, "V").Interior.Color = blueColor Or ws.Cells(sourceRow, "V").Interior.Color = redColor
                cellValue = ws.Cells(sourceRow, "V").Value

                If uniqueValues.Count = 0 Or Not IsInCollection(cellValue, uniqueValues) Then
                    ws.Cells(greenRow, 22 + colOffset).Value = cellValue
                    uniqueValues.Add cellValue, CStr(cellValue)
                End If

                ws.Cells(sourceRow, "V").ClearContents

                sourceRow = sourceRow + 1
                colOffset = colOffset + 1

                If sourceRow > lastRow Then Exit Do
            Loop

            sourceRow = greenRow + 1
            Set uniqueValues = New Collection
```

Figure A1. *Cont.*

```
            End If
        Next sourceRow

        Application.ScreenUpdating = True

        MsgBox "Unique metrics transfer to the above green row is complete!"
End Sub

Function IsInCollection(val As Variant, col As Collection) As Boolean
        On Error Resume Next
        col.Item CStr(val)
        IsInCollection = (Err.Number = 0)
        On Error GoTo 0
End Function
```

Figure A1. Transfer values in Column V across green cells across V to AA.

```
- Sub ChangeTextIfBlueBelow()
        Dim ws As Worksheet
        Set ws = ThisWorkbook.Sheets("Edited Version of ACN Data (2)")

        Dim blueRGB As Long
        blueRGB = [YourBlueRGB]

        Dim cell As Range, cellBelow As Range
        For Each cell In ws.Columns("AK").Cells
                Set cellBelow = cell.Offset(1, 0) ' Cell directly below
                If cellBelow.Interior.Color = blueRGB Then
                        cell.Value = "Yes"
                End If
        Next cell
End Sub
```

Figure A2. Changes values in Column AK.

```
Sub TransferData()
        Dim ws As Worksheet
        Set ws = ThisWorkbook.Sheets("Backup Version ACN Data 2")

        Dim sourceRow As Long, targetRow As Long, lastRow As Long
        Dim greenColor As Long, blueColor As Long, redColor As Long
        greenColor = 11854022
        blueColor = 15917529
        redColor = 11389944
```

Figure A3. *Cont.*

```
    lastRow = ws.Cells(ws.Rows.Count, "I").End(xlUp).Row

    Application.ScreenUpdating = False

    For sourceRow = 19 To lastRow
        If sourceRow > ws.Rows.Count Then Exit For

        If ws.Cells(sourceRow, "I").Interior.Color = greenColor And _
            ws.Cells(sourceRow + 1, "I").Interior.Color = blueColor And _
            ws.Cells(sourceRow + 2, "I").Interior.Color = redColor Then

            ' Transfer data from red cells to the green row across I, J, and K
            ws.Cells(sourceRow, "I").Value = ws.Cells(sourceRow + 2, "I").Value
            If sourceRow + 3 <= lastRow Then
                ws.Cells(sourceRow, "J").Value = ws.Cells(sourceRow + 3, "I").Value
            End If
            If sourceRow + 4 <= lastRow Then
                ws.Cells(sourceRow, "K").Value = ws.Cells(sourceRow + 4, "I").Value
            End If

            sourceRow = sourceRow + 2
        End If
    Next sourceRow

    Application.ScreenUpdating = True

    MsgBox "Data transfer is complete!"
End Sub
```

Figure A3. Transfers cells in Column I to green cells across I-K.

```
Sub TransferCargoTypeToAboveGreen()
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets("Good ACN Data 1")

    Dim sourceRow As Long, lastRow As Long
    Dim blueColor As Long, greenColor As Long
    greenColor = 11854022
    blueColor = 15917529

    lastRow = ws.Cells(ws.Rows.Count, "M").End(xlUp).Row ' Last row with data in column M

    Application.ScreenUpdating = False ' Turn off screen updating for performance
```

Figure A4. *Cont.*

```
    For sourceRow = 26 To lastRow
        ' Check if the current cell (sourceRow, "M") is blue
        If ws.Cells(sourceRow, "M").Interior.Color = blueColor Then
            ' Check if the cell directly above (sourceRow - 1, "M") is green
            If ws.Cells(sourceRow - 1, "M").Interior.Color = greenColor Then
                ' Transfer the cargo type from the blue cell to the green cell directly above
                ws.Cells(sourceRow - 1, "M").Value = ws.Cells(sourceRow, "M").Value
                ' Optionally clear the original blue cell after transfer
                ws.Cells(sourceRow, "M").ClearContents
            End If
        End If
    Next sourceRow

    Application.ScreenUpdating = True

    MsgBox "Cargo type transfer to above green cells is complete!"
End Sub
```

Figure A4. Transfers blue colored cells in Column M to green cells above.

Appendix B

Interest in improving the data analysis process arose following the automated scripts process, which, although automated, still required significant time for anomaly detection and data cleaning. As a result, neural network techniques using the Python code in Figure A5 were applied for data cleaning with the original dataset. A neural network model was created using freely available open-source resources. A preliminary algorithm was developed that used color coding to highlight duplicative and incomplete data. The results were then reviewed by a human subject matter expert. The algorithm was then updated using a second set of training data from the original dataset, and was run again on the large dataset. Redundant data were deleted and a tracer record was created for audit and compliance testing and analysis. Missing and incomplete data were flagged by the neural net so that the dataset could updated with the missing information. Machine learning was used to propose potential data to complete the data records, based on contextual information and data in the dataset. Process improvements in terms of time differences were recorded, comparing the use of the automated scripts and the neural nets algorithm. The Python code developed is shown in Figure A5.

```
import pandas as pd
from openpyxl import load_workbook
from openpyxl.styles import PatternFill

file_path = '/Users/jimmy/Desktop/ACN/Alaska Chadux Data, 11568 records, 12 December 2023.xlsx'
df = pd.read_excel(file_path)

highlighted_df = df.copy()

duplicate_mask = df.duplicated(subset=['Transit', 'Full Date+Time Entered EEZ', 'Full Date+Time Exited EEZ'],
keep='first')

incomplete_data_mask = df.loc[:, df.columns[32:37]].isnull().all(axis=1)

output_path = '/Users/jimmy/Desktop/ACN/Highlighted ACN Database.xlsx'
highlighted_df.to_excel(output_path, index=False, engine='openpyxl')

wb = load_workbook(output_path)
ws = wb.active

yellow_fill = PatternFill(start_color='FFFF00', end_color='FFFF00', fill_type='solid')
blue_fill = PatternFill(start_color='ADD8E6', end_color='ADD8E6', fill_type='solid')

for row_idx, is_duplicate in enumerate(duplicate_mask, start=2):
    if is_duplicate:
        for cell in ws[row_idx]:
            cell.fill = yellow_fill


for row_idx, is_incomplete in enumerate(incomplete_data_mask, start=2):
    if is_incomplete:
        for cell in ws[row_idx]:
            cell.fill = blue_fill

wb.save(output_path)

print("Data highlighting completed and saved to 'Highlighted ACN Database.xlsx'")
```

Figure A5. *Cont.*

```
output_path = '/Users/jimmy/Desktop/ACN/Highlighted ACN Database.xlsx'
highlighted_df.to_excel(output_path, index=False, engine='openpyxl')

wb = load_workbook(output_path)
ws = wb.active

yellow_fill = PatternFill(start_color='FFFF00', end_color='FFFF00', fill_type='solid')
blue_fill = PatternFill(start_color='ADD8E6', end_color='ADD8E6', fill_type='solid')

for row_idx, is_duplicate in enumerate(duplicate_mask, start=2):
    if is_duplicate:
        for cell in ws[row_idx]:
            cell.fill = yellow_fill

for row_idx, is_incomplete in enumerate(incomplete_data_mask, start=2):
    if is_incomplete:
        for cell in ws[row_idx]:
            cell.fill = blue_fill

wb.save(output_path)
```

Figure A5. Neural Network model for data cleaning.

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